Statistics practice problems : Ranging from basic to advanced

1. Descriptive Statistics

1. Calculate the mean, median, and mode of the following data set: [4, 8, 15, 16, 23, 42].
2. Find the range, variance, and standard deviation for the data set: [10, 12, 13, 16, 18, 22].
3. Given a frequency distribution, calculate the mean and standard deviation.
4. Construct a box plot for the data set [3, 7, 8, 5, 12, 14, 21, 13, 18].
5. Interpret the skewness and kurtosis of a given data set.
6. Calculate the interquartile range (IQR) of the data: [6, 7, 8, 15, 18, 22, 24, 26].
7. Find the z-score for a value of 70 in a distribution with a mean of 60 and standard deviation of 5.
8. Given a histogram, identify the mode and median.
9. Calculate the 95th percentile of the data: [55, 48, 71, 90, 61, 72, 80, 65].
10. Construct a stem-and-leaf plot for the given data.

2. Probability

1. What is the probability of rolling a sum of 7 with two dice?
2. A bag contains 4 red, 3 blue, and 5 green balls. What is the probability of drawing a red or a green ball?
3. What is the probability of drawing two aces from a deck of cards (without replacement)?
4. In a group of 10 people, what is the probability that two people share the same birthday?

5. A coin is flipped 3 times. What is the probability of getting exactly two heads?
6. Find the probability of drawing a heart or a queen from a deck of 52 cards.
7. Two dice are rolled. What is the probability of rolling a 4 or 5 on the first die and an even number on the second die?
8. If a student answers 5 multiple-choice questions randomly (each with 4 options), what is the probability of getting exactly 3 correct answers?
9. A factory has 3 machines. Machine A produces 30% of the items, B produces 50%, and C produces 20%. If an item is selected at random and found to be defective, what is the probability it was produced by Machine B (given B produces 2% defective items, A 3%, and C 1%)?
10. What is the probability of drawing 2 red cards and 1 black card from a deck of cards (with replacement)?

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3. Probability Distributions

1. Find the probability that a normally distributed random variable X with mean 50 and standard deviation 5 takes a value between 45 and 55.
2. A binomial random variable X has a probability of success of 0.4 in 6 trials. Find $P(X = 3)$.
3. For a Poisson-distributed variable with a mean of 4, find $P(X = 2)$.
4. Find the expected value and variance of a binomial distribution with $n = 10$ and $p = 0.3$.
5. A random variable X follows a geometric distribution with a probability of success 0.2. Find $P(X = 3)$.
6. Calculate the mean and variance of an exponential distribution with a rate parameter $\lambda = 2$.
7. Find the cumulative probability of a standard normal distribution up to $z = 1.96$.
8. In a Poisson distribution with $\lambda = 5$, find the probability that $X \geq 3$.
9. Given a normal distribution with a mean of 100 and standard deviation of 15, find the probability of X being between 85 and 115.

10. A continuous uniform distribution is defined on the interval [0, 10]. Find P(X < 3).

4. Hypothesis Testing

1. Conduct a one-sample t-test to determine if the mean score of a sample is significantly different from 50 (given sample data).
2. Perform a two-sample t-test to compare the means of two independent samples.
3. Conduct a paired t-test for before-and-after treatment data.
4. Test whether a coin is fair using a hypothesis test.
5. Conduct a chi-square goodness-of-fit test to determine if a die is fair (given observed frequencies).
6. Perform a chi-square test of independence for a given contingency table.
7. Conduct an ANOVA test to determine if there are significant differences between three groups' means.
8. Test the correlation coefficient between two variables for significance.
9. Perform a Wilcoxon rank-sum test for two independent samples.
10. Conduct a test for the variance of a population using the chi-square distribution.

5. Regression Analysis

1. Fit a linear regression model to the given data set and interpret the slope and intercept.
2. Calculate the coefficient of determination (R^2) for a linear regression model.
3. Conduct a hypothesis test on the slope of the regression line.
4. Perform multiple regression analysis on a data set with more than one predictor variable.
5. Given a logistic regression model, interpret the coefficients and predict the probability of an event.
6. Calculate the residuals for a fitted linear regression model and check for heteroscedasticity.

7. Test for multicollinearity using the Variance Inflation Factor (VIF).
8. Conduct polynomial regression for a quadratic relationship between the variables.
9. Use stepwise regression to select significant variables for a predictive model.
10. Perform ridge regression to handle multicollinearity in a linear regression model.

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6. ANOVA and Chi-Square

1. Perform a one-way ANOVA to test for differences among the means of three groups.
2. Conduct a two-way ANOVA to test the effects of two factors on a response variable.
3. Perform a Kruskal-Wallis test for non-parametric ANOVA.
4. Conduct a chi-square test for independence between two categorical variables.
5. Use a chi-square goodness-of-fit test to see if a distribution follows a normal distribution.
6. Test the homogeneity of variances using Levene's test in ANOVA.
7. Perform a Friedman test to analyze differences in rankings among multiple groups.
8. Conduct a Cochran's Q test for binary outcomes across multiple groups.

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7. Time Series Analysis

1. Plot the autocorrelation function (ACF) for a time series data set and interpret the lags.
2. Perform a decomposition of a time series into trend, seasonality, and residual components.
3. Fit an ARIMA model to a time series data set and interpret the parameters.
4. Test for stationarity using the Augmented Dickey-Fuller (ADF) test.
5. Calculate and plot the moving average of a time series.

6. Fit an exponential smoothing model to a time series and make predictions.
7. Use seasonal decomposition of time series (STL) for forecasting.
8. Perform a Ljung-Box test to check for autocorrelation in a time series.
9. Forecast future values using the Holt-Winters method.
10. Conduct a Durbin-Watson test for autocorrelation in residuals.

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8. Advanced Topics

1. Apply the central limit theorem to a problem involving sample means.
2. Perform a Bayesian analysis for a problem involving prior and posterior probabilities.
3. Derive the likelihood function for a binomial distribution and perform Maximum Likelihood Estimation (MLE).
4. Use the Bootstrap method to estimate the confidence interval for a sample mean.
5. Conduct a Monte Carlo simulation to estimate the probability of a certain outcome.
6. Apply the Markov Chain Monte Carlo (MCMC) method for a problem involving Bayesian inference.
7. Use principal component analysis (PCA) for dimensionality reduction in a high-dimensional data set.
8. Perform a cluster analysis (e.g., K-means clustering) on a data set and interpret the results.
9. Use logistic regression to classify binary outcomes.
10. Perform a survival analysis using the Kaplan-Meier estimator.
11. Conduct a Cox proportional hazards model for survival data.
12. Apply the Kolmogorov-Smirnov test to compare two distributions.

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9. Sampling and Confidence Intervals

1. Find the confidence interval for the mean of a normal population with known variance.
2. Construct a 95% confidence interval for a population proportion.

3. Calculate the sample size required for estimating a mean with a margin of error of 2.
4. Use stratified sampling to estimate the mean of a population.
5. Construct a confidence interval for the difference between two population means.
6. Perform cluster sampling on a given population.
7. Determine the confidence interval for a binomial proportion.
8. Calculate the confidence interval for the variance of a normal population.
9. Estimate the confidence interval for a population mean using a bootstrap method.
10. Use systematic sampling to collect a sample from a population.

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10. Non-Parametric Tests

1. Conduct a Mann-Whitney U test to compare two independent samples.
2. Perform a Spearman rank correlation test for two variables.
3. Use the Kolmogorov-Smirnov test for testing the goodness-of-fit.
4. Perform a Wilcoxon signed-rank test for paired samples.
5. Conduct a Shapiro-Wilk test for normality of the data.
6. Apply the Kruskal-Wallis test for comparing more than two groups.
7. Perform a rank correlation test using Kendall's

Tau.
98. Conduct a sign test for matched pairs.
99. Perform a Friedman test for repeated measures.
100. Use the Smirnov test for comparing two empirical distributions.

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11. Data Analysis and Interpretation

1. Analyze a given data set and choose the appropriate statistical test.
2. Interpret the p-value and confidence interval from a t-test.
3. Explain the meaning of statistical power and its implications for hypothesis testing.

4. Interpret the output of a multiple regression analysis.
5. Analyze a contingency table and perform a chi-square test of independence.

These problems will give a broad and deep understanding of statistics, from foundational concepts to advanced analysis.